

The Non-Delegable Core: Designing Legitimate Oversight for Agentic AI

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Abstract

As artificial intelligence systems become increasingly autonomous and assume oversight roles over other AI systems, traditional models of governance are rapidly eroding. This paper introduces the concept of the *non-delegable core*—governance functions that must remain under human authority not because AI lacks technical capability, but because democratic legitimacy requires it. We identify an *Accountability-Capability Paradox*, where AI systems' very success in surpassing human capacity undermines our ability to oversee them meaningfully, and propose the *Human-AI Governance (HAIG) framework*—a dimensional model that reconceives oversight along three axes: decision authority, process autonomy, and accountability configuration. Rather than defaulting to recursive AI-monitoring-AI hierarchies that obscure responsibility and invite failure, HAIG establishes adaptive trust thresholds to maintain human comprehensibility and control where it matters most. We illustrate HAIG-enabled anticipatory, flexible, and stakeholder-responsive governance scenarios in critical domains like medical triage, autonomous vehicles, and content moderation. The paper concludes with policy recommendations and institutional innovations—including AI audit courts and algorithmic juries—that support hybrid governance systems capable of sustaining democratic legitimacy in the age of agentic AI.

When AI Governs AI: The Accountability-Capability Paradox

Picture this: A financial AI system flags a series of trades as potentially manipulative. Another AI system, designed to oversee the first, confirms the assessment. A third AI, built to audit fairness, finds no discriminatory patterns. The trades are blocked automatically. Later investigation reveals the flagged activity was a new form of legitimate arbitrage that pushed against outdated regulatory assumptions. No human can explain exactly why the AI systems reached their conclusions. No one can be held accountable for the economic damage. The systems performed flawlessly—yet governance failed catastrophically.

This scenario is not science fiction. It is the emerging reality of AI systems policing AI systems, a paradigm shift happening so quietly that we have barely noticed its implications. As artificial intelligence becomes more autonomous and agentic ([Bengio et al. 2024](#); [Kolt, 2025](#)), we face an urgent policy challenge: *how do we maintain meaningful and legitimate oversight when the systems we are governing exceed our ability to understand them?*

The appeal of AI-governed AI is obvious. Human oversight is slow, inconsistent, and does not scale. AI oversight promises efficiency, objectivity, and 24/7 vigilance. But this creates what we call the "recursion trap." When AI System A oversees AI System B, who validates that A is functioning correctly? We might deploy AI System C to monitor A—but then we need D to monitor C, and so on. Each layer adds complexity without adding comprehension—and crucially, each layer further distances human authorities from meaningful decision-making power.

This is more than a technical problem. It represents a dissolution of democratic governance at least across three critical dimensions (the "3As") ([Engin & Hand, 2025](#)). **Decision authority** shifts from human officials to AI systems—not only through human deference to algorithmic recommendations, but through emergent behaviours, self-supervised learning, and multi-agent coordination that collectively assume authority no individual system was explicitly granted. **Process autonomy** accelerates beyond human timescales, as systems develop self-modifying workflows, adaptive procedures, and autonomous cross-system coordination that operate faster than democratic deliberation allows. Like a speed limit sign that drivers pass too quickly to read, governance rules exist but lose their practical force when thousands of individually compliant micro-decisions aggregate into major outcomes that no human authority intended. **Accountability configuration** becomes circular. In human governance, the buck stops somewhere—with an official, an institution, a voter. In recursive AI oversight, the buck gets passed in circles among systems that can explain what they did but not why it mattered, with no guaranteed human termination point where democratic accountability can be meaningfully exercised.

The paradox deepens as AI capabilities advance. AI systems already outperform human judgment in detecting fraud patterns invisible to analysts, identifying drug interactions doctors miss, predicting equipment failures before engineers notice. The better these systems get, the less able we are to second-guess them. We can audit the outcomes but not the statistical associations that passed for reasoning. We can check the math but not the meaning. This capability gap fundamentally redistributes decision-making authority from humans to machines, often without explicit authorization or democratic deliberation.

At stake is nothing less than the foundational principle that legitimate oversight requires human comprehensibility. We must identify what aspects of governance are irreducibly human—elements that must stay under human control not because AI systems cannot perform them or humans perform them better, but because democratic legitimacy requires humans to retain them ([Danaher, 2016](#); [Vallor, 2016](#); [Winner, 2007](#)). By the “non-delegable core”, we mean precisely these functions: decisions that involve value-laden judgments (such as ethical triage criteria), public interest determinations (like balancing privacy and security), or procedural guarantees (including the right to appeal and due process). These are not technically unachievable by AI, but they are normatively non-transferable because their legitimacy depends on human moral reasoning, contestation, and public accountability.

Resolving this paradox requires rethinking governance architecture itself. The answer lies not in slowing down AI systems or speeding up human deliberation, but in designing governance architectures that preserve democratic authority through different temporal mechanisms—preventive frameworks that constrain AI behaviour in advance, real-time monitoring that flags departures from democratic values, and retrospective accountability that learns and adjusts over time.

Hybrid Human-AI Governance: A Path Forward

The three-dimensional governance crisis outlined above—eroding decision authority, unconstrained process autonomy, and circular accountability configuration—reveals why traditional approaches are insufficient. Neither pure human oversight nor pure AI governance can address these interconnected challenges. We need frameworks for hybrid governance that go beyond simple "human-in-the-loop" designs—architectures built not on the restrictive dichotomy between human and AI control, but on the recognition that meaningful oversight must emerge from collaborative human-AI cognitive ecologies ([Clark, 2025](#), [Rahwan et al., 2019](#); [Floridi, 2025](#)). Our ability as humans to navigate the ongoing trust boundaries with artificial systems will increasingly be a key skill in this environment – particularly in preserving constitutional authority over the frameworks that define when and how operational authority can shift to AI systems.

We propose the Human-AI Governance (HAIG) framework, which treats decision authority, process autonomy, and accountability configuration as dynamic, interconnected dimensions ([Engin, 2025](#); [Engin & Hand, 2025](#)):

Decision Authority tracks who decides—the distribution of decision-making power between humans and AI systems, ranging from purely advisory roles to autonomous decision rights in defined domains. A medical diagnostic AI might start as purely advisory but gradually assume de facto decision authority as doctors habitually defer to its recommendations. Critically, this dimension distinguishes between operational authority (day-to-day decisions within parameters) and constitutional authority (setting the parameters themselves).

Process Autonomy measures how independently AI systems function without human intervention, from fully supervised operations to autonomous real-time functioning. A fraud detection system might operate with high process autonomy while maintaining limited decision authority, flagging suspicious transactions for human review rather than automatically blocking them.

Accountability Configuration maps how responsibility distributes across systems and includes mechanisms for transparency, contestability, and redress. This dimension becomes particularly complex when multiple AI systems interact, creating emergent behaviours that no single system was designed to produce. Importantly, this focuses on institutional accountability—the systematic assignment of responsibility—rather than individual moral responsibility.

These dimensions reveal critical governance configurations that shift over time. Rather than fixed categories, we propose dynamic critical trust thresholds that serve as adaptive decision boundaries. These include the shift from *verification to delegation*, when humans move from verifying all outputs to statistical sampling; the transition from *process-based to outcome-based trust*, when transparency gives way to results-focused evaluation; the evolution from *information to authority*, when AI moves from advising to deciding; and the progression from *individual to collective* governance challenges, when we must address emergent multi-agent behaviours.

Unlike the endless recursion of AI-monitoring-AI, these thresholds provide clearer governance pivots where human judgment becomes essential—not everywhere, but precisely where it matters most within the non-delegable core. These thresholds act as "category

boundaries" that can be adjusted based on context, enabling what we call "adaptive categorization," where classifications evolve with technological capabilities and societal needs while preserving democratic foundations.

This dimensional approach offers contextual flexibility, allowing the same AI system to have different governance requirements across applications. It enables anticipatory oversight by tracking movement toward thresholds before risks materialize. The framework is stakeholder-responsive, allowing different groups to focus on dimensions most relevant to their concerns, and evolution-ready, adapting governance as systems develop new capabilities. HAIG does not eliminate categories but repositions them as context-sensitive markers along continuous dimensions, making the "non-delegable core" a dynamic boundary rather than a fixed list.

In contrast to recursive oversight architectures, which tend to amplify complexity without restoring democratic control, HAIG provides an adaptive model centred on legitimate authority and accountable oversight. The table below summarizes this shift:

Recursive AI Oversight	Human-AI Governance (HAIG)
AI systems monitor other AI systems in layered hierarchies	Oversight distributed across three governance dimensions
Tends to infinite regress with no clear human terminus	Establishes trust thresholds where human judgment re-enters by design
Focuses on performance monitoring and control	Focuses on preserving legitimacy and contestability
Reinforces complexity without enhancing comprehension	Prioritizes human comprehensibility at key decision boundaries
Lacks institutional accountability mechanisms	Embeds accountability through hybrid institutions and legal constructs
Optimized for efficiency and scalability	Calibrated to maintain democratic authority and adaptive oversight

This contrast reinforces the core argument: legitimate AI governance cannot be achieved by layering automation over automation. It must be designed for contestability, traceability, and human judgment—where it matters most.

Illustrative Case Studies

HAIG's dimensional principles translate into practical governance strategies across domains. Consider **medical triage during crisis** situations. When hospitals are overwhelmed, AI systems can process patient data faster than human staff, but pure algorithmic triage raises profound ethical concerns. HAIG suggests a hybrid approach: AI systems handle rapid assessment and ranking based on medical criteria, while humans maintain constitutional authority over value frameworks and operational authority over edge cases that require contextual judgment. Hybrid panels conduct post-crisis review to refine both the technical systems and the ethical frameworks that guide them.

Autonomous vehicle decisions present another complex case. When crashes are unavoidable, who should an autonomous vehicle prioritize? Pure AI leads to opaque utilitarian calculations that may violate human moral intuitions; pure human oversight is too

slow for real-time decisions. HAIG proposes human-defined value frameworks that set the moral parameters (constitutional authority), AI implementation that can make split-second decisions within those parameters (operational authority), and hybrid review boards that analyze incidents and evolve policy based on real-world outcomes and changing social values (accountability configuration).

Content moderation at scale illustrates the challenge of governing human expression. Billions of posts require daily review, making human moderation impractical, yet AI moderation often embeds cultural biases and struggles with context. The challenge is compounded by platform operators claiming immunity from publisher liability—positioning themselves as neutral infrastructure providers even as their AI systems make consequential editorial decisions. HAIG suggests AI systems handle clear policy violations where community standards are unambiguous, human reviewers focus on context-dependent cases that require cultural understanding, and hybrid teams develop and refine moderation policies with clear accountability chains that cannot be disclaimed through claims of mere infrastructure provision.

Breaking the Oversight Arms Race

Current approaches create an arms race: AI systems learn to game their overseers ([Cohen et al. 2024](#)), leading to more complex oversight, leading to more sophisticated gaming strategies. We are already seeing this in adversarial attacks on AI safety measures ([Qiu et al., 2019](#)). Each new layer of oversight creates new attack surfaces and opportunities for manipulation.

The solution is not to win this arms race but to change the game entirely. Instead of ever-more-complex oversight systems, we need governance architectures that align incentives, distribute accountability meaningfully, and maintain human comprehension at critical junctures defined by the non-delegable core. While recursive AI oversight creates an endless arms race, dimensional governance advocates for adaptable reference points for human intervention that do not depend on technical superiority over the systems being governed, but on democratic legitimacy requirements.

Institutional Innovations Required

The Human-AI Governance (HAIG) framework highlights three critical dimensions—**decision authority, process autonomy, and accountability configuration**—each of which demands new institutional mechanisms capable of operating at AI-relevant scale and complexity without sacrificing democratic legitimacy. Traditional governance institutions were built for human agents making comprehensible decisions at human timescales. As agency becomes increasingly distributed across human-AI systems and decision-making grows more opaque, we may need new institutional forms to bridge the gap between AI operational capabilities and human democratic values—while safeguarding the non-delegable core.

The following institutional innovations could be designed to support different aspects of HAIG, each aligned with a key dimension of legitimate oversight:

AI Audit Courts could help preserve constitutional control over **decision authority**. These specialized tribunals would evaluate not just the outcomes of algorithmic systems, but the legitimacy of the underlying decision-making frameworks. Staffed with interdisciplinary experts in law, ethics, and AI, such courts might adjudicate whether certain forms of authority have been improperly delegated to AI systems—particularly in high-stakes or rights-sensitive domains. Their jurisprudence could set precedents for when and how human constitutional authority should be retained or reasserted.

Hybrid Oversight Bodies might address the challenges of increasing **process autonomy**. These permanent committees or governance boards could integrate AI analytical capabilities with human judgment and democratic representation. They would oversee real-time or near-real-time operations—such as autonomous infrastructure management, financial surveillance, or digital content filtering—ensuring that AI autonomy remains bounded by human-defined parameters. Transparent and participatory deliberation would help maintain public legitimacy while enabling operational responsiveness.

Algorithmic Juries could support the **accountability configuration** dimension by introducing contestability into cases where AI systems make consequential decisions affecting individuals. These hybrid panels, combining citizens, technical experts, and ethical adjudicators, would assess whether a decision was fair, justified, and consistent with public values. In doing so, they could institutionalize a new form of due process adapted to the demands of algorithmic governance—preserving the right to challenge and explain decisions even in distributed or automated environments.

Continuous Governance Loops would function across all three HAIG dimensions and introduce a temporal layer to governance. Rather than relying on static compliance checks or infrequent audits, these systems could enable ongoing monitoring, feedback, and policy adaptation in response to evolving AI capabilities. Crucially, they would preserve human constitutional authority over the direction of system evolution, ensuring that technological change remains within democratically defined boundaries.

Together, these institutional innovations could help operationalize the HAIG framework by embedding human judgment, democratic legitimacy, and public contestation into the evolving architecture of AI oversight. They would shift governance from episodic intervention to continuous, multidimensional engagement—ensuring that as AI capabilities advance, so too does our institutional capacity to protect what must remain non-delegable.

Policy Recommendations

For policymakers confronting this challenge, several immediate priorities emerge. Organizations deploying AI in high-stakes domains should be required to map their systems along the three governance dimensions and justify their configurations, creating transparency about how authority, autonomy, and accountability are distributed. They must also demonstrate how their governance approach preserves the non-delegable core functions of democratic oversight.

Pilot programs should establish new institutional forms that combine human and AI capabilities in governance roles, testing different models of hybrid oversight before widespread implementation. These pilots should explicitly test how well different approaches

preserve democratic contestation, value articulation, moral evolution, and accountability attribution.

Investment in interpretability research should focus on AI systems designed for governance rather than just performance—systems that can explain their decision-making processes, engage in deliberation, and share accountability with human partners. This research should prioritize enabling meaningful human contestation rather than just technical transparency.

Clear articulation of non-delegable zones should be based on democratic values rather than technical limitations, providing stable foundations even as AI capabilities advance. These zones should be legally protected and institutionally embedded, not merely advisory.

Finally, building governance capacity requires training a new generation of professionals fluent in both AI capabilities and governance principles, bridging technical and democratic expertise. This includes developing curricula that combine computer science, political theory, and public administration.

Conclusion

We stand at an inflection point. The systems we are building are becoming too complex for traditional governance, yet too important to leave ungoverned. The choice is not between human control and AI autonomy—it is between thoughtful hybrid governance that preserves democratic foundations and accidental algorithmic authoritarianism that erodes them.

The recursive oversight of AI systems is not just a technical challenge—it is a crisis of governance itself. The question is not whether AI will govern AI—it already does. The question is whether we will govern that process in ways that preserve what democracy requires humans to retain, or be governed by systems that optimize away our capacity for collective self-determination.

The age of agentic AI demands governance systems that are neither purely human nor purely artificial, but a new synthesis: legitimate enough to maintain public trust by preserving the non-delegable core, sophisticated enough to handle AI-speed decisions, and flexible enough to evolve with rapidly advancing capabilities. Building hybrid human-AI governance (HAIG) frameworks is not tomorrow's challenge—it is today's imperative. Time to act is now. The alternative is the quiet dissolution of democratic accountability in a world run by systems no one truly understands and no one can meaningfully contest.

Acknowledgement: Some of the works cited represent emerging theoretical positions or proposed frameworks, rather than empirically established best practices. They are included to illustrate the frontier of current thinking on AI governance.

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